

Please make sure that you show all of the steps, not just the answer.

1) Find the equation of the line that passes through (2,8) and (4,-6).

• Slope $m = \frac{y_2 - y_1}{x_2 - x_1}$ $m = \frac{-6 - 8}{4 - 2} = \frac{-14}{2} = -7$

• Point-Slope form:

$$y - y_1 = m(x - x_1)$$

$$y - 8 = -7(x - 2)$$

• Slope intercept form:

$$y - 8 = -7x + 14$$

$$y = mx + b$$

$$\boxed{y = -7x + 22}$$

2) Simplify each expression completely.

a) $\frac{(x+3)^2 - 3^2}{x}$

$$\frac{(x^2 + 6x + 9) - (9)}{x} =$$

$$\frac{x^2 + 6x}{x} = \frac{x(x+6)}{x} =$$

$$\boxed{x+6}$$

b) $\frac{x}{\sqrt{x+3} - \sqrt{3}}$

$$\frac{x}{\sqrt{x+3} - \sqrt{3}} \cdot \frac{\sqrt{x+3} + \sqrt{3}}{\sqrt{x+3} + \sqrt{3}} =$$

$$\frac{x(\sqrt{x+3} + \sqrt{3})}{(\sqrt{x+3} - \sqrt{3})(\sqrt{x+3} + \sqrt{3})} =$$

Difference of Squares

$$\frac{x(\sqrt{x+3} + \sqrt{3})}{(\sqrt{x+3})^2 - (\sqrt{3})^2} =$$

$$\frac{x(\sqrt{x+3} + \sqrt{3})}{(x+3) - (3)} =$$

$$\frac{x(\sqrt{x+3} + \sqrt{3})}{x} =$$

$$\boxed{\sqrt{x+3} + \sqrt{3}}$$

3) Solve each equation.

a) $\log(x+1) + \log(2x) = 1$

$$\log a + \log b = \log(ab)$$

"Addition rule"

$$\log((x+1)(2x)) = 1$$

$$\log(2x^2 + 2x) = 1$$

Rewrite to exponential form

$$10 = 2x^2 + 2x$$

$$2x^2 + 2x - 10 = 0$$

$$2(x^2 + x - 5) = 0$$

$$x^2 + x - 5 = 0 \text{ (quadratic)}$$

$$\rightarrow x = \frac{-1 \pm \sqrt{1^2 + 4(1)(5)}}{2}$$

$$x = \frac{-1 \pm \sqrt{21}}{2}$$

$$\text{Reject } x = \frac{-1 - \sqrt{21}}{2}$$

Due to domain Restrictions of logs. $\boxed{x = \frac{-1 + \sqrt{21}}{2}}$

b) $3e^{6x} - 2 = 13$

$$3e^{6x} = 15$$

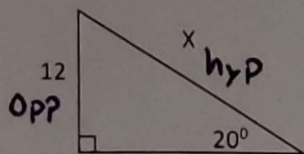
$$e^{6x} = 5$$

$$\ln(e^{6x}) = \ln(5)$$

$$6x = \ln(5)$$

$$\boxed{x = \frac{\ln(5)}{6}}$$

4) Solve for the variable, X.

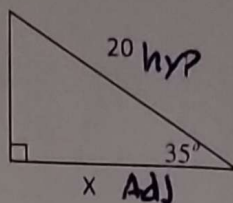


$$\sin = \frac{\text{Opp}}{\text{Hyp}}$$

$$\sin(20^\circ) = \frac{12}{X}$$

$$X \sin(20^\circ) = 12$$

$$X = \frac{12}{\sin(20^\circ)} \approx \frac{12}{0.3420} \approx 35.1$$



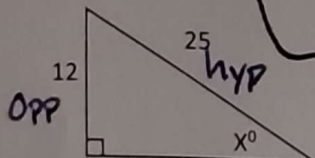
$$\cos = \frac{\text{Adj}}{\text{Hyp}}$$

$$\cos(35^\circ) = \frac{X}{20}$$

$$X = 20 \cos(35^\circ)$$

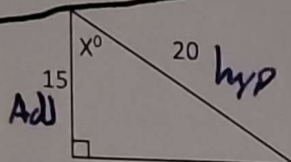
$$X \approx 20(0.8192)$$

$$X \approx 16.4$$



$$\sin(X) = \frac{12}{25}$$

$$X = \sin^{-1}\left(\frac{12}{25}\right) \approx 28.7^\circ$$



$$\cos(X) = \frac{15}{20}$$

$$\cos(X) = \frac{3}{4}$$

$$X = \cos^{-1}\left(\frac{3}{4}\right) \approx 41.4^\circ$$

5) Find all solutions in the interval $[0, 2\pi)$. Give exact answers like $\frac{2\pi}{7}$

a) $8 + 8\cos(x) = 12$

b) $2\sin(2x) - 1 = 0$

$$\frac{8\cos(x)}{8} = \frac{4}{8}$$

$$\cos(x) = \frac{1}{2}$$

cos is pos in QI and QIV

↳ the reference angle is $\frac{\pi}{3}$

$$x = \frac{\pi}{3}, \quad x = 2\pi - \frac{\pi}{3}$$

$$x = \frac{5\pi}{3}$$

$$2\sin(2x) = 1$$

$$\sin(2x) = \frac{1}{2}$$

sub θ for $2x$

$$\sin(\theta) = \frac{1}{2}$$

Reference angle is $\frac{\pi}{6}$

and sin is Positive in QI and QII so:

$$2x = \frac{\pi}{6}, \quad 2x = \frac{5\pi}{6}$$

$$x = \frac{\pi}{12}, \quad x = \frac{5\pi}{12}$$

Then add π to each for the next cycle in $[0, 2\pi)$

$$x = \frac{13\pi}{12}, \quad x = \frac{17\pi}{12}$$